

Final report of library management system

Abstract

This report represents a simple library management system implemented in c programming language. The program allows users (students and teachers) to login with valid ID and search for a book using the unique book number assigned by UET Abbottabad. Once found it displays book's name, book's author, and shelf number. The code displays basic data handling, conditional logic, and searching in arrays using C structures.

Introduction

Managing books in library is a common but important task in academic institutes. This project involves creating a small-scale library management system using C. The goal is to store information of about 60 books and allow authenticated users to search for a specific book using its number.

The program verifies user identity based on ID (S for student and T for teachers). Once authenticated, the user inputs a UET Abbottabad assigned book number, and the program searches through the stored data. If the book is found, its details are displayed.

This project showcases how structures and arrays can be used to manage and access complex data in C.

C Program

```
#include <stdio.h>

#include <stdlib.h>

#define max_books 60

struct Book {
    int book_num;
    char name[50];
    char author [50];
    char shelf[10];
};

int main()
{
```

```

struct Book books [max_books]={

{1309,"Intel micro_processor","Barrey b.brey and C.R Sharma","4"},

{4125,"Advanced engineering mathematics","Ervin kreyszig","4"},

{9033,"Elements of eloectromagnetics","Mathio.N.O.Sadiko","4"},

{4303,"Introduction to computer","Peter nortoris","12"},

{4341,"Calculus and analytical geometry","Thomos finney","12"},

{5341,"Introdution to VLSI cicuit and system","John P uyemura","10"},

{1771,"Electronic devices","Thomas L.floyd","10"},

{5441,"Modern digital and analog communication system","B.P lathi and zhi
dhing","10"},

{873,"Semi-conductor devices","Jim fry","10"},

{3313,"Physics of electrical engineering","Anthony johm pointen","10"},

{3499,"Signals and systems","Alan v.oppenthum.alan s.willsky and s.hamid
nawab","10"},

{1721,"Digital logic and computer design","M.Morris mano","11"},

{7799,"HUGHES Electrical and electronic technology","edward hughes","11"},

{3126,"Wireless communications","eheodr s.rappaport","11"},

{6583,"ETHICS IN ENGINEERING","mike w.mortan","11"},

{2403,"Engineering electromagnetics","Kenneth R.demaresr","14"},

{704,"Engineering electromagnetics","William H.Hayt.JR","14"},

{3871,"Engineering mechanics statics","J.L meriam.L.G Kraige","14"},

{2932,"Engineering economy","Leland blank.anthony tarquin","14"},

{2399,"Engineering electromagnetics","Kenneth R.demaresr","14"},

{708,"Engineering electromagnetics","William H.Hayt.JR","14"},

{3867,"Engineering mechanics statics","J.L meriam.L.G Kraige","14"},

{1506,"Engineering economy","E.Paul degarmo william G.sullivan james
A.bontabelli","14"},

{2407,"Engineering economy","Leland blank.anthony tarquin","14"},

{703,"Engineering electromagnetics","William H.Hayt.JR","14"},

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{2936,"Engineering electromagnetics","Kenneth R.demaresr","14"},

{1475,"Engineering economy","E.Paul degarmo william G.sullivan james A.bontabelli","14"},

{4170,"Mathematical methods","S.M. Yusuf,abdul majeed,muhammad amin","20"},

{5079,"Power electronics","Cyril w.lander","20"},

{2840,"Modern control sysytems","Richard c.dorf,robert h.bishop","20"},

{1584,"data structures using c and C++2","Yedidyah langsam , moshe j.augenstein.aaron m.tenenbaum","20"},

{3769,"engineering mechanics dynamics","J.L.meriam.L.G. kraige","20"},

{4186,"Mathematical methods","S.M. Yusuf,abdul majeed,muhammad amin","20"},

{2811,"Modern control sysytems","Richard c.dorf,robert h.bishop","20"},

{5057,"Power electronics","Cyril w.lander","20"},

{1605,"data structures using c and C++2","Yedidyah langsam , moshe j.augenstein.aaron m.tenenbaum","20"},

{2500,"Introduction to statistical theory part 1","PROF. sher muhammad chaudhary,DR. shahid kamal","3"},

{2057,"Object-Oriented program in C++","Robert lafore","3"},

{4525,"Introduction to computers","peter nortans","3"},

{5119,"Differential equations","Dennis G.zinn,micheal R.cullen","3"},

{2987,"probability,random variables and stochkstic processes","Athanasios papoulis,s.unnikrishhna pillai","3"},

{122,"engineering mechanics dynamics","J.L.meriam.L.G. kraige","20"},

{1660,"complex variables","murray R. spiegel","3"},

{2998,"probability,random variables and stochkstic processes","Athanasios papoulis,s.unnikrishhna pillai","3"},

{5102,"Differential equations","Dennis G.zinn,micheal R.cullen","3"},

{2482,"Introduction to statistical theory part 1","PROF. sher muhammad chaudhary,DR. shahid kamal","3"},

{1683,"complex variables","murray R. spiegel","3"},

{3001,"probability,random variables and stochkstic processes","Athanasios papoulis,s.unnikrishhna pillai","3"},

```

{5155,"Differential equations","Dennis G.zinn,micheal R.cullen","3"},
{3849,"electronic devices and circuits","Theodore F.bogart JR","1"},
{90,"introductory electronic devices and circuits","Robert T.paynter","1"},
{4227,"introductory electronic devices and circuits","Robert T.paynter","1"},
{86,"introductory electronic devices and circuits","Robert T.paynter","1"},
{64,"introductory electronic devices and circuits conventional low version","Robert
T.paynter","1"},

{3826,"electronic devices and circuits","Theodore F.bogart JR","1"},
{0347,"Introdution to VLSI cicuit and system","John P uyemura","10"},
{0515,"control system engineering ", " Norman S.nise","10"},

{3399,"Signals and systems","Alan v.oppenenthum.alan s.willsky and s.hamid
nawab","10"},

{5376,"Modern digital and analog communication system","B.P lathi and zhi
dhing","10"},

{1725,"Digital logic and computer design","M.Morris mano","11"},

```

```

};

char id[20];

printf("enter your id (starting with S or T):");

scanf("%s",id);

if(id[0]!='S' &&id[0]!='T') {

    printf ("invalid login,login failed");

    return 0;

}

printf("login successful");

int bookNum;

printf("enter the UET assigned book_number");

scanf("%d",&bookNum);

int found = 0;

for (int i=0;i<max_books;i++){

```

```

if (books[i].book_num==bookNum){
    printf("book found");
    printf("book name : %s\n ", books[i].name);
    printf("book author : %s\n", books[i].author);
    printf("book shelf number :%s\n", books[i].shelf);
    found=1;
    break;
}
}
if(!found){
    printf("this book is not found", bookNum);
}
return 0;
}

```

Conclusion

This program is the basic representation of how real-world systems like library databases can be simulated using C language. Through the use of arrays, structures, and control statements, the system enables user authentication and book lookup features. Although it uses static dataset, the concept can be expanded to include file handling or database integration for real-world application.

References:

- GeeksforGeeks: <https://www.geeksforgeeks.org/structures-in-c/>
- TutorialsPoint: <https://www.tutorialspoint.com/cprogramming/index.htm>
- Textbooks used for book data (engineering standard references)